



Plant oil-derived copolymers with remarkable post-polymerization induced mechanical enhancement for high performance coating applications

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HIGHLIGHTS

- High oleic soybean oil (HOSO) derived copolymers were prepared using a semi-batch emulsion polymerization method.
- Successful polymerization of soybean monomer with various co-monomers resulted in latexes with tailorable properties.
- A simple auto-oxidative crosslinking of latex films provided excellent mechanical enhancements.

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ABSTRACT

Polymer coatings have been heavily utilized in many industrial and civil applications where a protective thermoset material with enhanced properties such as chemical and physical resistance, thermal stability, and tailorable mechanical properties is desired. Plant oils are a class of promising biomass toward sustainable polymers, but have yet to be fully harnessed in tailorable thermoset coatings due to the challenging functionalization and extensive crosslinking processes required to achieve desirable thermomechanical properties. In this work, we demonstrated that soybean methacrylate (SBMA) from high oleic soybean oil (HOSO) can be utilized to produce bio-based acrylic thermoset copolymers through an industrially viable semi-batch emulsion polymerization process with various commonly used co-monomers such as methyl methacrylate, styrene, and butyl acrylate. A wide range of monomer feed ratios with SBMA from 0 to 50 wt% was easily achieved with minimal modifications allowing for good tunability of thermal and mechanical properties in the prepared latexes. More importantly, a simple and effective auto-oxidative crosslinking of the latex films provided extreme mechanical enhancements making these thermosets good candidates in ultra-strong, ultra-tough, and high T_g coating applications.

1. Introduction

Thermoset polymers have been widely used as high performance coatings in many industrial applications due to their thermal stability, chemical and mechanical resistance, facile processing, and tailorable mechanical properties. Chemically and mechanically resistant thermoset polymers are frequently used as specialized coatings on substrates for heavy duty work such as laboratory benches, automotive parts, and construction materials. Resins, including epoxy and phenol, are widely used as durable coating materials due to their low cost and facile crosslinking chemistry [1,2]. Unfortunately, the curing agents, solvents, and additives in thermoset resin materials often pose health hazards, such as known carcinogenic formaldehyde or suspected endocrine disruptor bisphenol A [3]. Extensive efforts have been focused

on replacing these hazardous components with benign bio-derived alternatives [4–10]. On the other hand, acrylate-based thermoset systems present an alternative with better tailorable properties, owing to their greater structural diversity and safer options for post-polymerization functionalization [11]. Acrylate monomers of various structures and chemical compositions can be copolymerized with a variety of co-monomers including styrenes, vinyl ethers, and other (meth)acrylates [12]. Higher molecular weight and better chemical composition control can be achieved in comparison to the epoxy and phenolic resin-based systems as the initial polymerization and subsequent crosslinking occur in two separate steps. Furthermore, crosslinking in these systems can be conveniently achieved by utilizing light and/or heat without the need to use toxic catalysts or other additives [13–15]. However, these acrylate-based materials have relied primarily on petrochemicals raising

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